

Stylized Facts about AI Capabilities

Tom Cunningham, METR

Today

Forecasts/ Timelines

Substantial disagreement, mostly about the future path of AI capabilities.

How People use ChatGPT

Biggest fact: the primary impact of LLMs has been for home production.

Stylized Facts About AI Capabilities

Understanding AI capabilities is intellectual quicksand. Five facts and their implications for modelling AI (work with Ali Merali).

METR Time Horizon

This is the best proxy we have for measuring AI capabilities, but it's still unsatisfactory.

Bay Area Tech-Econ Seminars

**Twenty six
seminars since
2022**

1,000 people on mailing list.

Year	Spring Seminars	Fall Seminars
2022	0	4
2023	5	3
2024	5	2
2025	3	4

Organizers

- Stacy Carlson (Amazon)
- Tom Cunningham (METR)
- Andrew Hobbs (USF)
- Guido Imbens (Stanford)
- Peter Lorentzen (USF)
- Rose Tan (Stanford)
- Sean Taylor (OpenAI)

Thanks: Laura Ventura, Elizabeth Kuhn Wilsey.

Forecasts of AI & Growth, 2025-2035

date of forecast	author	annual excess growth, 2025-2035	quote
March 2023	Briggs and Kodrant (2023) (Goldman Sachs)	+15%	"We estimate that widespread adoption of generative AI could raise overall labor productivity growth by around 1.5pp/year (vs. a recent 1.5% average growth pace), roughly the same-sized boost that followed the emergence of prior transformative technologies like the electric motor and personal computer."
May 2023	Baily, Brynjolfsson, and Korinek (2023)	+2.8%	"The projection labeled 'Level' assumes that generative AI raises the level of productivity and output by an additional 18% over ten years, as suggested by the illustrative numbers we discussed for the first channel. After ten years, growth reverts to the baseline rate. The third projection labeled 'Level+Growth' additionally include a one percentage point boost in the rate of growth over the baseline rate, resulting from the additional innovation triggered by generative AI."
June 2023	McKinsey Global Institute (2023)	+0.1-0.6%	"GenAI alone: +0.1-0.6 pp/yr labor productivity through ~2040; combined with other re/automation: +0.5-3.4 pp/yr"
August 2023	Tyler Cowen	+0.25%-0.5%	"My best guess, and I do stress that word guess, is that advanced artificial intelligence will boost the annual US growth rate by one-quarter to one-half of a percentage point."
March 2024	Korinek and Suh (2024)	+18%	"In the baseline AGI scenario ... steady-state growth of 18% per year."
April 2024	Acemoglu (2024)	+0.07%	"Using existing estimates on exposure to AI and productivity improvements at the task level, these macroeconomic effects appear nominal but modest—no more than a 0.71% increase in total factor productivity over 10 years."
May 2024	Alderson et al. (2024) (BIS)	+2.5%	"They assume a 1.5% growth in productivity but then predict that equilibrium output will increase at a higher rate. "We assume that AI raises annual productivity growth by 1.5 percentage points for the next decade, in line with plausible estimates in the literature ... Growth is fastest in the first 10 years – i.e. the period in which AI directly raises industry-level TFP – at which point GDP is almost 30% higher than it would have been without"
June 2024	Aghion and Burel (2024)	+0.68-1.3%	"Based on the first approach, we estimate that the AI revolution should increase aggregate productivity growth by between 0.8 and 1.3pp per year over the next decade. Using the second approach but with our own reading of the recent empirical literature on the various components of the task-based formula, we obtain a median estimate of 0.68pp additional annual total factor productivity (TFP) growth."
July 2024	Tyeb (2024)	0.2-0.9%	Exhibit 1 shows a range of estimates for increases to productivity growth rates, between 0.2% and 0.9%.
Dec 2024	Filippucci, Gal, and Schiffré (2024) (OECD)	+0.25-0.6%	"Main estimates for annual aggregate total-factor productivity growth due to AI range between 0.25-0.6 percentage points (0.4-0.9 pp. for labour productivity)."
Feb 2025	Tyler Cowen	+0.5%	"I've gone on record as suggesting that AI will boost economic growth rates by half a percentage point a year."
March 2025	Bergstead et al. (2025) (ECB)	+0.29%	"We predict TFP gains of 2.9% in the medium run (say in the next ten years) in the euro area, equivalent to an additional 0.29 percentage points per year."
March 2025	Endl et al. (2025) (Epoch GATE model)	+30%	The website figures showing Gross World Product shows that in 2035 "default" path hits approximately a 30% annualized growth rate. However Ege Endl says: "I don't personally predict 30% mean annual job growth in the US over the next 10 years ... the model does with some reasonable parameter values, but my hunches for that kind of growth are longer, and there's model uncertainty and so on."
April 2025	Mouch et al. (2025) (IMF)	+0.2%	"We find that the medium-term productivity gains for Europe as a whole are likely to be modest, at around 1 percent cumulatively over five years."
May 2025	Jack Clark (Anthropic)	+3-5%	"I think my bear case on all of this is 3 percent, and my bull case is something like 5 percent."
June 2025	Filippucci et al. (2025) (OECD)	+0.4-1.3%	"annual aggregate labour productivity growth due to AI range between 0.4-1.3 percentage points in countries with high AI exposure ... In contrast, the estimated range is 0.2 to 0.8 percentage points in countries where these determinants of AI gains are less favourable (e.g. Italy, Japan)."

Forecasts of AI & Growth, 2025-2035

Most economists' forecasts are 0.1–1.5%/year.

Notable exceptions Baily, Brynjolfsson, and Korinek (2023), Korinek and Suh (2024).

Most AI insiders' forecasts are 3–30%/year.

Notable exception: Andrej Karpathy says he expects GDP growth to remain on historical trends.

We can decompose the disagreement into parts:

1. Capabilities (share of work automated, Hulten's theorem)
2. Speed of diffusion.
3. Degree of complementarity between AI-produced and human-produced goods.

The disagreement is mostly about capabilities growth.

Economists: fixed capabilities, gradual diffusion.

AI insiders: growing capabilities, immediate diffusion.

What we've seen: (1) capabilities doubling every year; (2) diffusion fast, but hard to quantify.

How People Use ChatGPT

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September 2025

We acknowledge help and comments from Joshua Achiam, Hemanth Asirvatham, Ryan Beiermeister, Rachel Brown, Cassandra Duchan Solis, Jason Kwon, Elliott Moksiki, Kevin Rao, Harrison Satcher, Gawesha Weeratunga, Hannah Wong, and Analytics & Insights team. We especially thank Tyna Eloundou and Pamela Mishkin who in several ways laid the foundation for this work. This study was approved by Harvard IRB (IRB25-0983). A repository containing all code run to produce the analyses in this paper is available on request. The views expressed herein are those of the authors and do not necessarily reflect the views of the National Bureau of Economic Research.

At least one co-author has disclosed additional relationships of potential relevance for this research. Further information is available online at <http://www.nber.org/papers/w34255>

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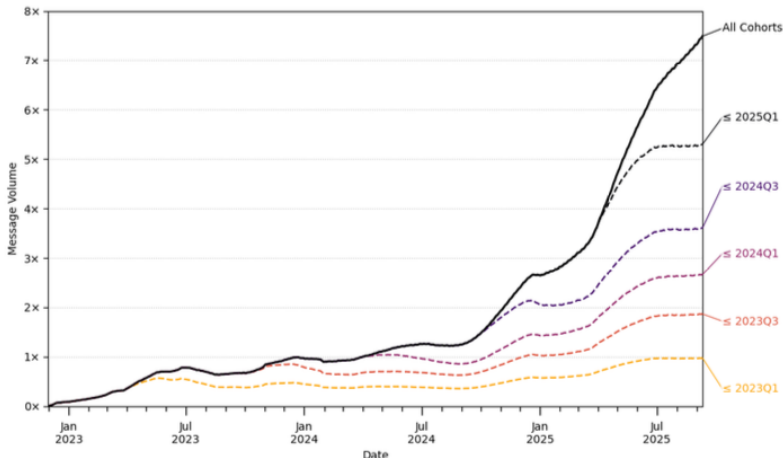
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How People Use ChatGPT

800M weekly ChatGPT users.

Growth in both extensive and intensive margins despite substantial increase in competition.

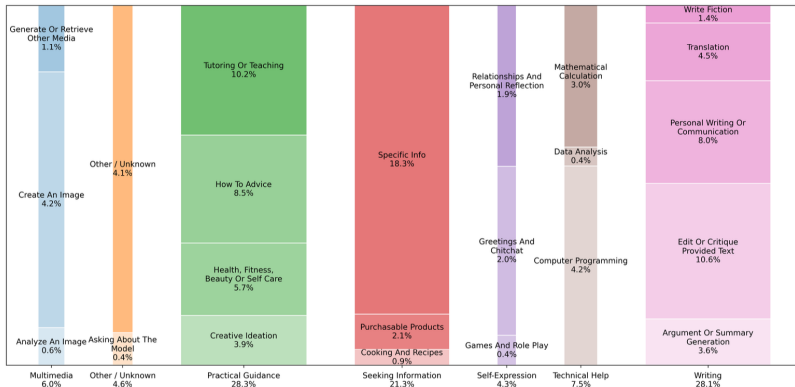
The volume of queries within-cohort more than doubled over mid-2024 to mid-2025. This is likely causal. Implication: value has doubled.



How People Use ChatGPT

Around 2/3 of messages are not work-related.

GenAI's primary impact is *home production*, solving everyday problems. Plausible that it's a substitute for market labor.



Stylized Facts about AI Capabilities

Genesis of this work.

In summer 2024 OpenAI was preparing to launch the first reasoning model (strawberry AKA o1-preview), I was asked to help estimate the likely economic impact.

The new model showed huge improvements on reasoning-intensive tests (MATH, GPQA), scores comparable to a PhD student.

What will be the impact of a model with PhD-level intelligence?

Ideal: plug a vector of capabilities into an economic model, & predict the change in equilibrium.

We don't have such a model.

Stylized Facts and Implications

Fact about AI	Implication for Economics of AI
1. We have no standard metric for computer intelligence	Opportunity for theorists.
2. LLM capabilities are highly correlated.	Directed technical change may be difficult.
3. LLM capabilities are growing rapidly.	Diffusion isn't the primary cause of growth.
4. LLM capabilities are distinct from human capabilities.	Need a two-factor model of ability.
5. There is no consensus on the bottleneck.	Opportunity for theorists.
6. Inference prices are falling at 10X+/year.	The compute share of GDP may be small.
7. LLMs can either do a task cheaply or not at all.	Equilibrium task-sharing seems unlikely.

We have no standard metric for computer intelligence.

A graveyard of measures of computer intelligence:

Measure	Interpretation	Passed Humans
Chess	Shannon: <i>“Chess is generally considered to require ‘thinking’ for skilful play.”</i>	1997
Wino-grad	<i>“when people pass the test, we would say they were thinking.”</i>	2019
ARC-AGI	<i>“the only formal benchmark of AGI progress.”</i>	2024
MMMU	<i>“Benchmark for Expert AGI.”</i>	2025
AGIEval	<i>“general abilities ... in tasks pertinent to human cognition”</i>	2025

There are some live proposals, none are canonical.

Most definitions are either (1) too narrow to generalize, or (2) too vague to measure.

- Morris et al. (2023) (Google DeepMind) five levels of AGI.
- OpenAI five levels of AGI, not officially released.
- Hendrycks et al. (2025) “the definition of AGI.”

This is a hard problem!

It’s intellectual quicksand, many smart people have got themselves stuck. Sensible to wear snowshoes.

Many Taxonomies Proposed, None Widely Adopted

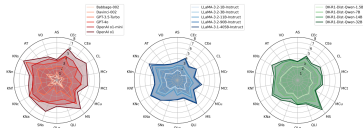


Figure 8: Ability profiles of the 15 LLMs. An ability of f means that there is 50% probability of the model to succeed on questions at demand level f (that leads to some abilities being beyond 5). In contrast to radial plots usually shown for LLMs in the literature (7, 51) the values shown here are actual abilities on a ratio scale (0, 100) and the values (in expectation) are more robust to changes in the difficulty distributions of the benchmarks used. In Figure 10, we additionally show the ability profiles for the 15 LLMs using the Abo-e-Light battery v1.0, which yields nearly the same profiles. In Appendix 9.2, we show clear scaling curves of model abilities as a function of Φ parameters.

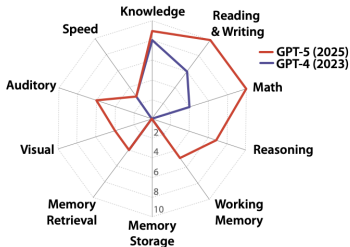
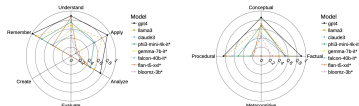


Figure 1: The capabilities of GPT-4 and GPT-5. Here GPT-5 answers questions in 'Auto' mode.

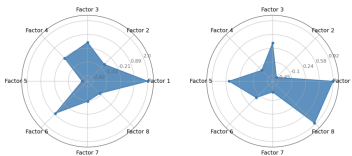
Table 2: Model performance on Bloom's Taxonomy's Cognitive Dimension and Knowledge Types. Model scores marked with * are based only on our own evaluation on BBH and AGIEval.



(a) Performance of models by cognitive process dimension

(b) Performance of models by knowledge type

Figure 2: Performance of models mapped to Bloom's Taxonomy. Model scores marked with * are based only on BBH and AGIEval. Unmarked scores are based on all benchmarks.



(a) Flan-T5-XL

(b) OpenChat-3.5

Figure 4: Projection of new models

What Metrics of AI Capabilities do we Have?

Two common metrics of AI capability.

1. **Scores on benchmarks.** There are hundreds of different benchmarks, but no canonical set, and most get saturated quickly. Continual scramble to create new benchmarks.
2. **User pairwise preference.** Given two answers to a query, which do you prefer? These are used by labs to measure user utility, also some 3rd-party data sources, esp. LMSys chatbot arena.

Other metrics of AI capability.

1. **Causal effect on user retention.** Used by labs but the results are not generally reported.
2. **Uplift effects on human productivity.** Expensive to collect, (a) very wide range of results; (b) hard to compare over time because subjects and situation constantly changing.
3. **User willingness to pay.** Few studies. Brynjolfsson and Collis report \$100/user/month.
4. **Adoption.** 1B+ users, and tens of billions in expenditure, but hard to infer area under demand curve.

LLM capabilities are growing rapidly

Benchmarks are falling rapidly.

It typically takes 18 months for LLM scores on a benchmark to go from 25% to 75%.

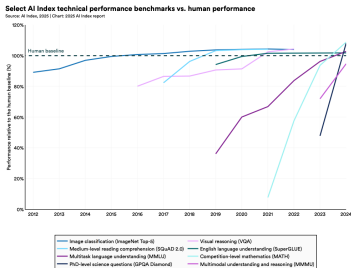


Figure 2.4.33

LLM capabilities are growing rapidly

**Latent ability
across benchmarks
is growing
continuously.**

Frontier model capabilities have grown steadily over time



Latent model ability (“Epoch capability index”).

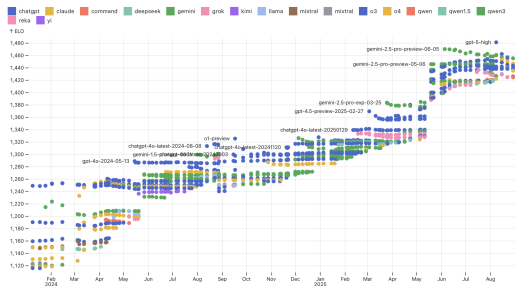
**Usage/user
doubled over
2024-2025.**

(see above)

LLM capabilities are growing rapidly

User preference scores are growing rapidly.

Models have 70% win-rate over 1yo models (150 Elo/year).



70% is high considering (1) noise; (2) heterogeneous preferences; (3) ceiling effects.

Implications of Capability Growth

Economic forecasts should incorporate capabilities growth. Argued above.

There are limits on the usefulness of RCTs.

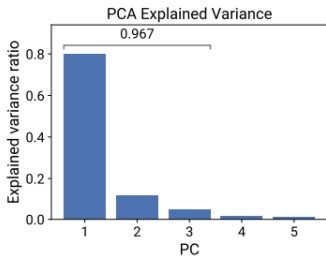
Measuring the effect with RCTs is not sufficient.

1. The causal effects are changing very rapidly.
2. We expect highly heterogenous effects.
3. We expect strong effects from experience.

LLM Capabilities are Highly Correlated

Many studies find high correlations across scores.

E.g. Ruan, Maddison, and Hashimoto (2024) finds the first factor explains 80% of variance:



(a) PCA explained variance

Additionally:

1. High correlation between benchmark scores and user-preference rankings (e.g. chatbot arena)
2. High correlation between benchmark scores and training compute.

LLM Capabilities are Highly Correlated

What does capability correlation mean concretely?

The model that is best at math is also probably best at programming, medical advice, legal advice, general knowledge, etc.

Earlier generations of AI instead showed *specialization*: DeepBlue was good at Chess, Watson was good at Jeopardy.

Supply-side explanation

Different investments route through the same latent quality:

$$q_1 = f_1(g(x_1, x_2))$$

$$q_2 = f_2(g(x_1, x_2))$$

Implicit model of much AI: scaling laws, transfer learning, bitter lesson.

Interpretation: larger models recover deeper latent structures, & thus are better at everything (“manifold hypothesis”, “platonic representation hypothesis”).

Demand-side explanations

Complementarity in demand: people want models that can do everything.

Hotelling competition: principle of minimum differentiation.

LLM Capabilities are Highly Correlated

Qualifications

1. There is some tradeoff across frontier capabilities, but a frontier model today typically dominates all frontier models from 6 months ago.
2. There is still a tradeoff between general capability and cost.
3. Post-training with RL may induce less correlation in abilities.

Implication: Directing Technical Change may be Hard

Some have proposed building AI to augment human labor.

Acemoglu (2021):

“our current trajectory automates work to an excessive degree while refusing to invest in human productivity.”

Brynjolfsson (2023):

“an excessive focus on developing and deploying [human-level AI] can lead us into a trap ... when AI is focused on augmenting humans rather than mimicking them, then humans retain the power to insist on a share of the value created”

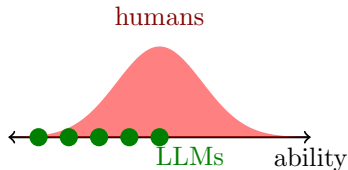
Our ability to direct change may be limited.

If there is truly a single latent factor of model ability, then work to train an *augmenting* model looks very similar to work to train an *automating* model, & the ability to steer AI capabilities may be limited.

LLM capabilities are distinct from human capabilities

We sometimes describe AI capability by human equivalent.

- years of education (Aschenbrenner (2024))
- percentile of human ability (Google, OAI)
- vertical knowledge (Ide and Talamas (2024))



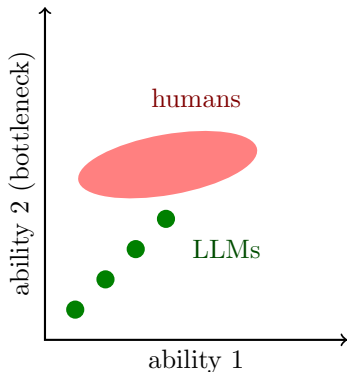
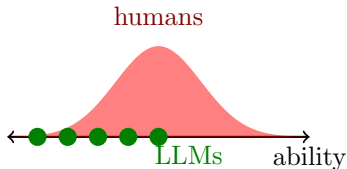
A single factor model is a poor fit.

LLMs are better than the best human at some tasks.
LLMs are worse than the worst human at other tasks.

Proxies for human ability are bad for LLM ability.

LLMs are able to answer almost all exam questions and interview questions for many jobs.
LLMs are not able to do the associated job.

There is a missing factor, we don't know what it is



There is no consensus on the bottleneck

Some candidate bottlenecks are already resolved.

When GPT-4 was released it could perform very well on exams, but the community quickly identified a variety of weaknesses and formalized them. Many of those bottlenecks have been resolved:

Capability	Benchmark	Apr 2023 (GPT-4)	Dec 2025
Deductive reasoning	MATH	23%	98%
	GPQA	36%	93%
Information-gathering	GAIA	15%	88%
Long context	Needle-in-Haystack	50% @ 128K	99.7% @ 1M
Accuracy / hallucination	SimpleQA Verified	~30%	72%
Multimodal inputs	MMMU-Pro	~35%	81%
In-context learning	ARC-AGI	7%	88%

Once a problem has been well-defined, LLMs have a very strong track record of solving it.

There is no consensus on the bottleneck

**Some candidate
bottlenecks at the
end of 2025:**

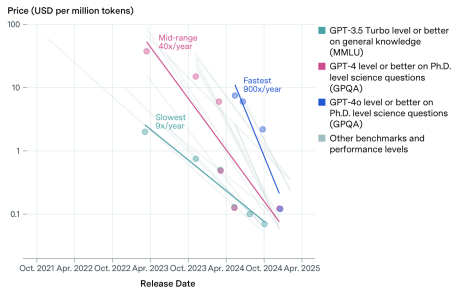
There is no consensus on what is the bottleneck:

Bottleneck	Status	Benchmark
Cost.	Does not seem an important bottleneck.	
Multimodal outputs	Still struggling	GDPVal
Novel situations	Mullainathan	ARC-AGI-2
Interactive decision-making	Still bad at playing games	BALROG, AI village, VendingBench
Hard-to-verify rewards	Karpathy	Human-graded rewards
Continual learning	Amodei, Karpathy, Hassabis	

Inference prices are falling at 10X/year

Enormous efficiency gains.

The price of a fixed level of capability has been falling at 10X+/year:



There is also growth in quality but it's harder to measure (time horizon, latent factor).

Why are prices falling?

Not because of lower margins – we see the same trend looking at raw compute cost for open models.

Smaller models are getting better through (1) better training techniques; (2) more training compute; (3) more training data; (4) better training data.

LLMs can either do a task cheaply or not at all

Ability to do tasks. METR (2024) say “when agents *can* do a task, they can often do so at a fraction of the cost of humans.”

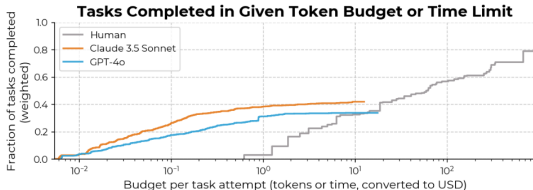


Figure 6: When agents can do a task, they can often do so at a fraction of the cost of humans. The diminishing marginal improvements in agent performance as a function of cost are evident here, contrasting with the log-linear relationship observed in our human baseliners.

Some exceptions to this rule.

1. Voice transcription for a brief period.
2. Some reasoning tasks (ARC-AGI), briefly.
3. Fixed costs (Svanberg et al. (2024)).

Implication: wages are unlikely to be pinned down by the price of compute

**Some models
feature
equilibrium
task-sharing:**

Humans perform tasks that computers are able to do, but it is too expensive for computers:

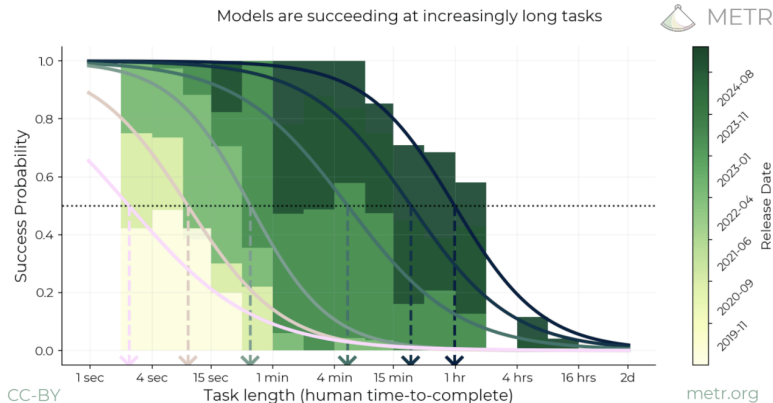
- Trammell and Korinek (2023)
- Ide and Talamas (2024)
- Smith (2024)
- Acemoglu (2024)

**Equilibrium
task-sharing would
require either:**

1. A dramatic increase in AI prices to perform a task:
 - An increase in the compute required for new tasks (the price of frontier inference has been steady).
 - An increase in compute costs (GPU rental prices have been steady, and the purchase/rent ratio has been stable).
 - An increase in margins.
2. A dramatic drop in human wages.

METR Time Horizon

Collect many *computer-using tasks*, and sort them by the time it takes a human to complete (the humans have relevant background but no context).



LLMs are consistently better at tasks that take humans less time. Can define a model's ability by its effective time horizon (50th percentile).

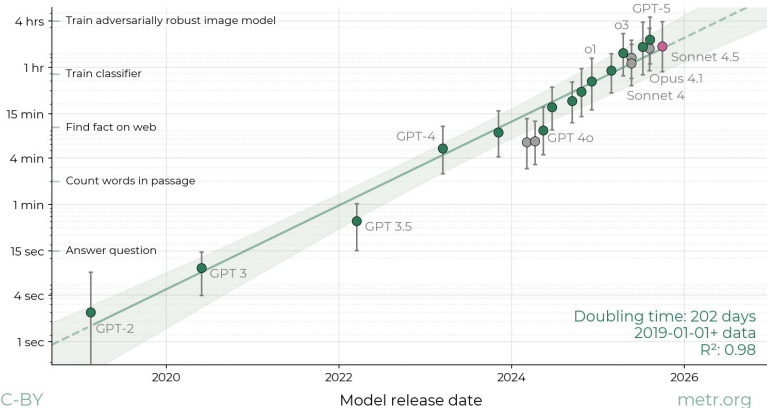
Time Horizon and Model Ability

Time horizon as a metric of model ability.

By this metric, we see a relatively smooth growth in time horizon over time.

One advantage of time horizon: solves the cold start problem. You can predict performance in advance.

Claude Sonnet 4.5 has a 50%-time-horizon of about 1 hr 53 min (95% CI: 50 - 235 min)
Task length (at 50% success rate)



Time Horizon Weaknesses

Tasks with long horizons but easy for LLMs.

E.g. find all the mentions of horses in Anna Karenina.

Tasks with short horizons but hard for LLMs.

Tasks with images, physical interaction.

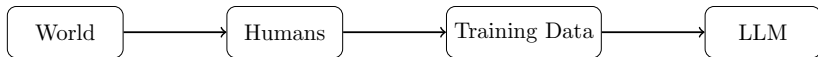
No definition of how tasks are selected.

We are running out of long tasks: difficult job of choosing more tasks, but remain faithful to the original distribution.

No theory of why long tasks are hard for LLMs.

Long tasks aren't much harder for humans, they just take longer. Long tasks must be different in some way, but unclear why.

What Would a Satisfying Theory Look Like?



Some implications:

1. LLMs are capped at human abilities.
2. LLMs are capped at training data.
3. Chatbots consulted on questions that are *new to you*: doctors use it for legal questions; lawyers use it for medical questions.
4. Chatbots relatively worse at off-distribution questions.

METR is Hiring

Ideal

1. Scrappy.
2. Self-sufficient with software engineering.
3. Familiar with AI capabilities & evals.
4. Ideas about using econ/statistics for understanding risks & capabilities.
5. Experience

Environment.

Pay is competitive with tech salaries (but no equity).

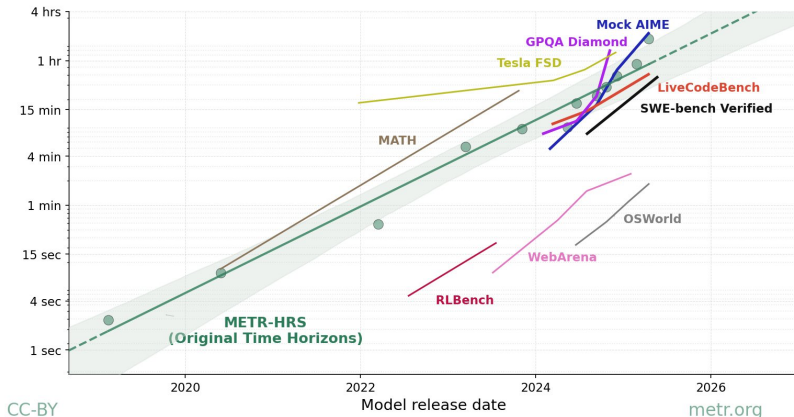
It feels like the center of the universe.

End

Model Evaluation and Threat Research (METR)

AI time horizons are increasing in many domains

Task length (at 50% success rate)

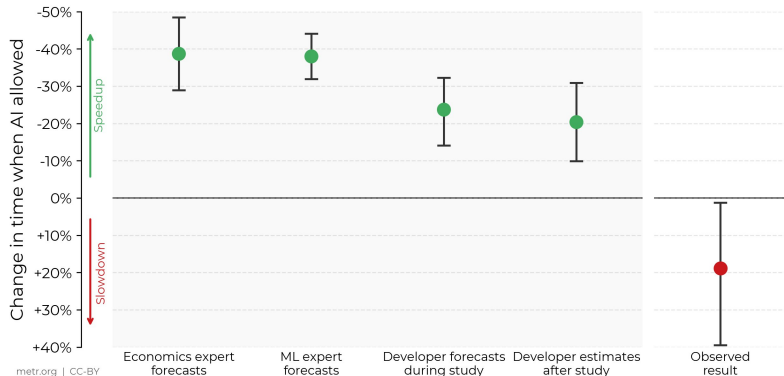


METR Uplift Study

Against Expert Forecasts and Developer Self-Reports, Early-2025 AI Slows Down Experienced Open-Source Developers



In this RCT, 16 developers with moderate AI experience complete 246 tasks in large and complex projects on which they have an average of 5 years of prior experience.



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